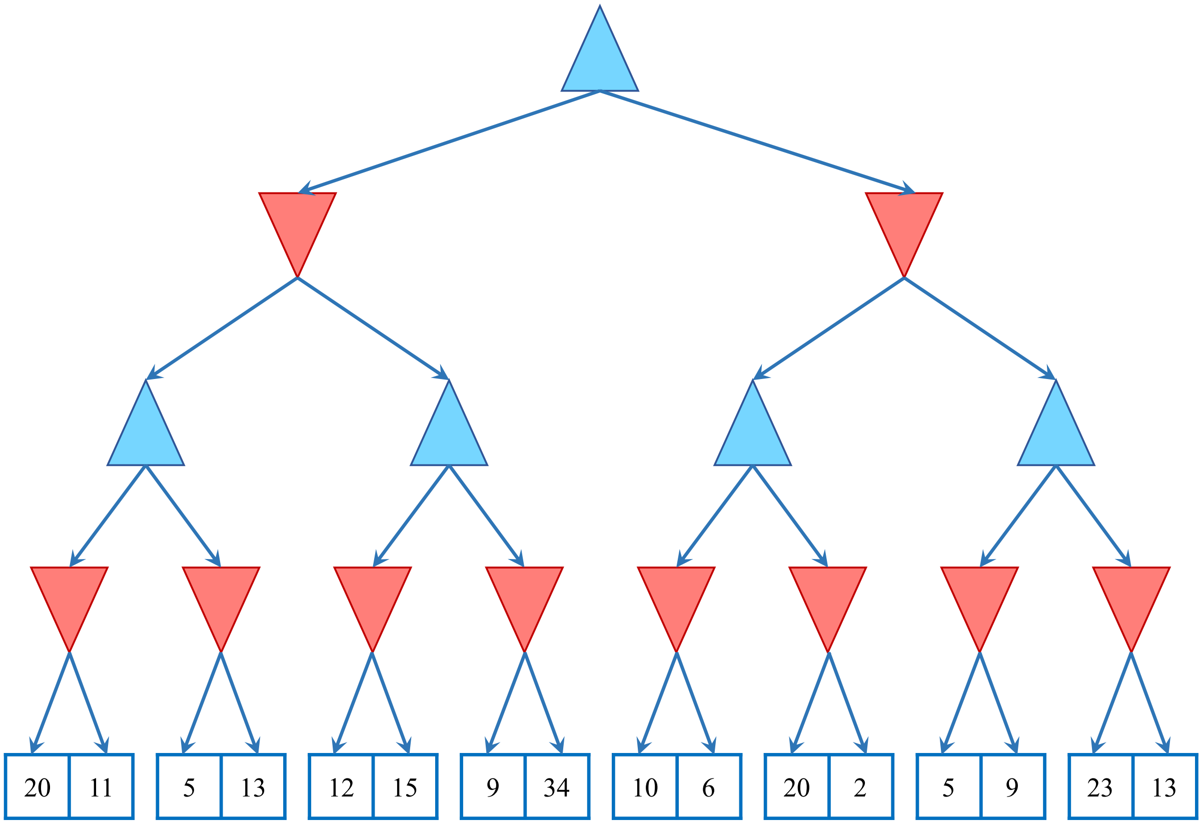
Adversarial Search

Consider the mini-max tree, whose root is a max node, shown below. Assume that children are explored left to right.



(a) (**3 pts**) Fill in the mini-max values for each of the nodes in the tree that aren't leaf nodes

(b) (**6 pts**) If α-β pruning were run on this tree, which branches would be cut? Mark the branches with a slash or a swirl (like a cut) and shade the leaf nodes that don't get explored.

Expectimax

(**6 pts**) You, are playing tic tac toe as the "X" player. You opponent, Olivia ("O" player) is a child, who is playing randomly. Since Olivia is very short, she is much more likely to fill "O"s into lower rows. Specifically, you know that Olivia is three times as likely to choose a space on the middle row than the top row, and three times again more likely to choose a space on the bottom row than the middle row. What move should you make to maximize your chance of winning? (Circle the board position and provide justifications)